GEN AI Developer Assessment

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SECTION A: Multiple Choice Questions

Time: 20 minutes | 40 Marks | 2 marks each

Google AI Studio (Q1-Q5)

Q1. In Google AI Studio, what is the primary purpose of "structured prompts"?

a) To generate images from text
Ans->b) To create reusable prompt templates with examples and variables
c) To connect to external databases
d) To deploy models to production

Q2. Which model family is available in Google AI Studio for text generation?

a) GPT-4
b) Claude
Ans->c) Gemini
d) LLaMA

Q3. What does the "temperature" parameter control in Google AI Studio?

a) Processing speed of the model
Ans->b) Randomness/creativity of the output
c) Maximum token limit
d) Model version selection

Q4. In Google AI Studio, what is a "freeform prompt"?

a) A prompt with predefined structure and examples
Ans->b) An open-ended prompt without structured format
c) A prompt that only accepts JSON input
d) A prompt connected to external APIs

Q5. What file format does Google AI Studio use when you export a prompt to code?

a) .txt only
Ans->b) .py (Python) and other language options
c) .exe
d) .prompt

Python FastAPI (Q6-Q10)

Q6. What is the correct way to define a POST endpoint in FastAPI?

a) @app.post("/items/")
b) @app.POST("/items/")
c) @post.app("/items/")
Ans->d) @route.post("/items/")

Q7. Which library does FastAPI use for data validation?

a) Marshmallow
b) Cerberus
Ans->c) Pydantic
d) Django REST Framework

Q8. What command runs a FastAPI application using Uvicorn?

a) python app.py
b) fastapi run app.py
Ans->c) uvicorn main:app --reload
d) flask run

Q9. In FastAPI, how do you define an optional query parameter?

a) def read_item(q: str):
Ans->b) def read_item(q: str = None):
c) def read_item(q: Optional[str]):
d) Both b and c are correct

Q10. What does FastAPI automatically generate at /docs?

a) Database documentation
Ans->b) Interactive Swagger UI API documentation
c) Source code
d) Error logs

LangChain (Q11-Q15)

Q11. What is the primary purpose of a "Chain" in LangChain?

a) To store data in a database
Ans->b) To sequence multiple LLM calls or operations together
c) To encrypt API keys
d) To manage user authentication

Q12. Which component in LangChain is used to split large documents into smaller chunks?

a) DocumentLoader
Ans->b) TextSplitter
c) VectorStore
d) Embedder

Q13. What is the role of RetrievalQA chain in LangChain?

a) To train new models
Ans->b) To retrieve relevant documents and answer questions based on them
c) To translate languages
d) To generate images

Q14. In LangChain, what does an "Embedding" represent?

a) A compressed version of the original text
Ans->b) A numerical vector representation of text
c) An encrypted form of data
d) A summary of the document

Q15. Which is a valid way to initialize ChatGoogleGenerativeAI in LangChain?

Ans->a) ChatGoogleGenerativeAI(model="gemini-pro")
b) GoogleChat(model="gemini")
c) GeminiLLM(version="pro")
d) LangChainGoogle(model="gemini")

LangGraph (Q16-Q20)

Q16. What is LangGraph primarily used for?

a) Creating static flowcharts
Ans->b) Building stateful, multi-actor applications with LLMs
c) Visualizing training data
d) Database schema design

Q17. In LangGraph, what is a "Node"?

a) A database connection
Ans->b) A function that performs a specific task in the graph
c) A user interface element
d) An API endpoint

Q18. What does the StateGraph class in LangGraph manage?

a) User sessions only

Ans->b) The flow and state of the application across nodes

c) Database transactions

d) File storage

Q19. How do you define conditional routing in LangGraph?

a) Using if-else in the node function only

Ans->b) Using add_conditional_edges() method

c) Using switch statements

d) Conditional routing is not supported

Q20. What is the purpose of END in LangGraph?

a) To terminate the Python program

Ans->b) To mark the final node where graph execution stops

c) To close database connections

d) To log errors

SECTION B: Case Study — Build a RAG Chatbot

SCENARIO: Customer Support Chatbot for TechGear Electronics

TechGear Electronics needs a customer support chatbot that answers product questions using a knowledge base. Build a RAG (Retrieval Augmented Generation) chatbot using ChromaDB, LangChain, and LangGraph.

📁 Knowledge Base (product_info.txt)

Product: SmartWatch Pro X Price: ₹15,999 | Features: Heart rate, GPS, 7-day battery, water resistant 50m Warranty: 1 year standard, 2 years extended (₹1,999) Product: Wireless Earbuds Elite Price: ₹4,999 | Features: ANC, 24-hour battery, Bluetooth 5.2 | Warranty: 6 months Product: Power Bank Ultra 20000mAh Price: ₹2,499 | Features: Fast charging 22.5W, USB-C & USB-A | Warranty: 1 year Return Policy: 7-day no-questions-asked. Refund in 5-7 business days. Support: Mon-Sat, 9AM-6PM IST | support@techgear.com

🎯 Tasks

Task 1: Setup & Document Loading [10 marks] → Load the knowledge base text file → Split into chunks using LangChain's TextSplitter → Create embeddings and store in ChromaDB

Task 2: RAG Chain Implementation [20 marks] → Create a retriever from ChromaDB → Build a RAG chain that retrieves context and generates answers → Use Google's Gemini model via LangChain

Task 3: LangGraph Workflow [20 marks]

→Node 1: Classifier — Categorize query (products / returns / general)

→ Node 2: RAG Responder — Use RAG to answer

→ Node 3: Escalation — Return escalation message for unhandled queries → Implement conditional routing based on classifier output

Task 4: FastAPI Endpoint [10 marks]

→ Create POST endpoint accepting user query

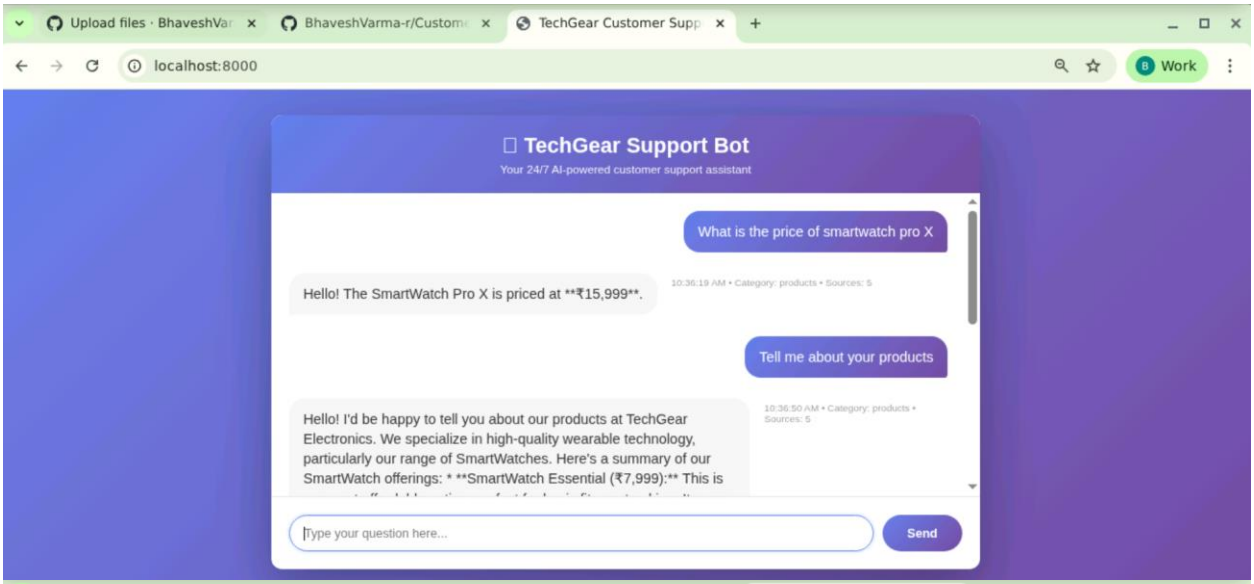
→ Process through LangGraph workflow

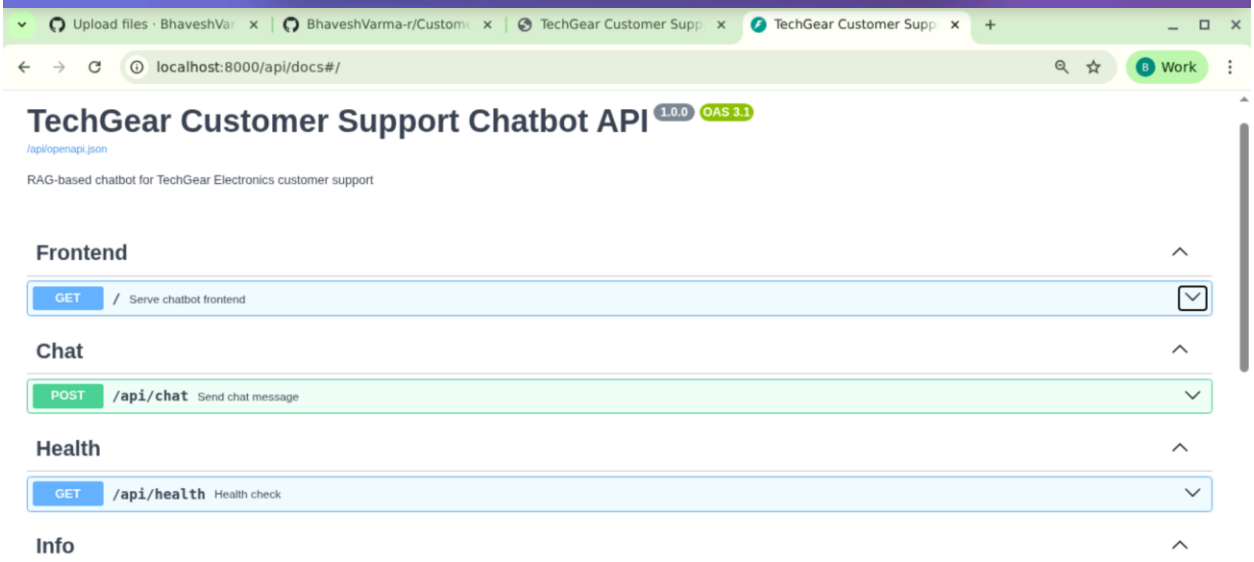
→ Return chatbot response as JSON

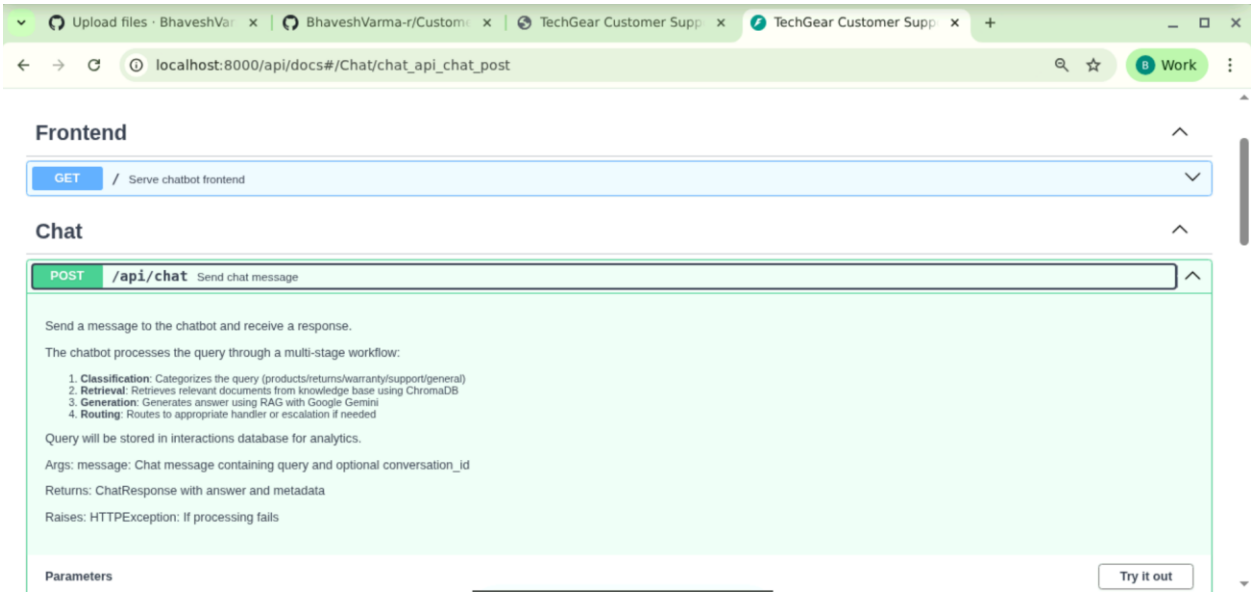
Solution:

Github Repo Link: https://github.com/BhaveshVarma-r/Customer_SupportBot

Screenshots:






Upload files · BhaveshVar
BhaveshVarma-r/Custom
TechGear Customer Supp
TechGear Customer Supp
localhost:8000/api/docs#/Chat/chat_api_chat_post
Work
Frontend
GET / Serve chatbot frontend
Chat
POST /api/chat Send chat message
Send a message to the chatbot and receive a response.
The chatbot processes the query through a multi-stage workflow:
1. Classification: Categorizes the query (products/returns/warranty/support/general)
2. Retrieval: Retrieves relevant documents from knowledge base using ChromaDB
3. Generation: Generates answer using RAG with Google Gemini
4. Routing: Routes to appropriate handler or escalation if needed
Query will be stored in interactions database for analytics.
Args: message: Chat message containing query and optional conversation_id
Returns: ChatResponse with answer and metadata
Raises: HTTPException: If processing fails
Parameters
Try it out